

Lec-04 MATRICES

$m \times n$ array of scalars

$$A = \begin{bmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \dots & a_{mn} \end{bmatrix}$$

m rows $\overline{a_1}$ n columns $\overline{a_n}$

$$\underbrace{\overline{a_1}, \overline{a_2}, \dots, \overline{a_n}}_{\text{columns}}$$

n vectors of size $m \times 1$

a_{23} = entry in 2nd row
3rd column

a_{ij} = entry / element in i^{th} row
 j^{th} column

row index $1 \leq i \leq m$
column index $1 \leq j \leq n$

Ex: $A = \begin{bmatrix} 3 & -1 & 2 \\ -2 & 4 & 3 \end{bmatrix}$

2 row 3 columns

2x3 matrix

$$a_{22} = 4$$

$$a_{23} = 3$$

$[A_{ij}] \rightarrow ij^{\text{th}}$ entry of matrix A .

$$a_{mn} \in \mathbb{R}$$

$$A \in \mathbb{R}^{m \times n}$$

A is real matrix

Set of $m \times n$ real matrices

$$a_{mn} \in \mathbb{C}$$

A is complex matrix

$$A \in \mathbb{C}^{m \times n}$$

Set of $m \times n$ complex matrices

Basic operations

Addition

A + B

We can add 2 matrices

A, B only if they
have same size

$$C = A + B$$

$$c_{ij} = a_{ij} + b_{ij}$$

$$\underbrace{\begin{bmatrix} 3 & -1 & 2 \\ -2 & 4 & 3 \end{bmatrix}}_A + \underbrace{\begin{bmatrix} 4 & -3 & +6 \\ 2 & 2 & -1 \end{bmatrix}}_B$$

$2 \times 3 \qquad \qquad \qquad 2 \times 3$

$$= \begin{bmatrix} 7 & -4 & 8 \\ 0 & 6 & 2 \end{bmatrix}$$

Scalar Multiplication.

$$[k \cdot A]_{ij} = k a_{ij}$$

↑ multiply each element
of A by scalar

$$2A = 2 \begin{bmatrix} 3 & -1 & 2 \\ -2 & 4 & 3 \end{bmatrix}$$

$$= \begin{bmatrix} 6 & -2 & 4 \\ -4 & 8 & 6 \end{bmatrix}$$

MATRIX MULTIPLICATION

$$\begin{matrix} A & B \\ m \times n & p \times q \end{matrix}$$

AB is defined only if $n = p$.

no. of columns
of A

no. of rows
of B

$$\begin{matrix} A \cdot B \\ m \times n & n \times q \end{matrix}$$

$$C = AB$$

$$C_{ij} = \sum_{k=1}^n a_{ik} b_{kj}$$

inner product of i^{th} row of A
with j^{th} column of B .

Ex: $A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}_{2 \times 3}$

$B = \begin{bmatrix} 1 & -2 \\ -1 & 1 \\ 1 & 3 \end{bmatrix}_{3 \times 2}$

AB is defined.

$AB = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix} \begin{bmatrix} 1 & -2 \\ -1 & 1 \\ 1 & 3 \end{bmatrix}$

Diagram illustrating the calculation of the product AB . The first row of A (1, 2, 3) is circled in orange and labeled $1,1$. The first column of B (1, -1, 1) is circled in orange and labeled $1,1$. The second column of B (-2, 1, 3) is circled in orange and labeled $1,2$. The second row of A (4, 5, 6) is circled in pink and labeled $2,1$. The second column of B (-2, 1, 3) is circled in pink and labeled $2,2$. The resulting product matrix is $\begin{bmatrix} 2 & 9 \\ 5 & 15 \end{bmatrix}$.

$m \times n$

$n \times q$

$AB: m \times q$

A

B

$\Rightarrow$

AB

2×3

3×2

2×2

i, j th element of AB is

inner product of i th row of A
 j th column of B .